SUSTAINABLE DEVELOPMENT
Exploring the contradictions

Nonita T. Yap

Sustainable Development Is

a concept whose time seems finally to have arrived in the social and economic agenda. The phrase itself has been around for years. Economists and developmenalists talked of sustainable development long before the publican of the *World Conservation -Strategy* in 1986 or Our *Common Future* in 1987. The use of the phrase seems to have gone full circle - from the economists to the environmentalists and now back to the economists the definition having undergone some change in the process.

In the current usage, sustainable development refers to the relationship of humans to their environment. The connections between environment and development are both complex and fascinating, but the discussions of the two issues have largely taken place in isolation from one another. The notion of sustainable development brings them together.

As a process, development involves a dynamic interaction between human systems and environmental systems.

The human system is encompassed by the environmental system (Fig. 1). Human systems include the various forms of human organizations, their technologies and the culture through which these interactions are played out.

The environmental system performs three functions essential in maintaining a particular quality of life: (1) as a supplier of resources, renewable and non-renewable, for the systems of production; (2) as a sink for the wastes of human activities; and, (3) as a source of life-support systems, specifically, air, water, soil and the assimilative capacities provided by various ecological processes. These three functions are the environmental constraints within which development takes place. People change the quality of life by using these environmental functions through technology and economic organization.

Technology can redress humanly defined environmental "shortcomings" but it may not be costless. Each technological intervention has the potential for creating conditions that limit the human capacity to sustain the system. The cost of this intervention, actual and perceived, varies from community to community, even with the same piece of technology.

Environmental change is in itself not a concern. The environment is constantly undergoing natural changes. The natural environment has a certain degree of resilience, of robustness. Ecologists speak of the "carrying capacity" of an ecosystem. i.e., the amount of life it can support: but they also speak of its "assimilative capacity", the capacity to absorb and recover from insults, the capacity to heal itself. Many environmental changes are cyclical or self-correcting.

Sustainable development raises concern over the environmental disturbance caused by human intervention and the speed with which the environmental system can adapt to restore the balance between human demands and environmental capacity. “To sustain” means “to keep alive”, “to endure”. It is worth stressing that both the human and environmental systems must adjust to each other. The speed of change affects the ability of both systems to adjust.

If the two systems cannot adjust in time, the end result may be a gap between human and environmental capacity, a gap that cannot be bridged, and a loss of sustainability.

The challenge of sustainable development may be pictured as one of achieving a balance between population and patterns of consumption on the one side and the mix of environmental functions on the other, with technology as a non-rigid spherical pivot and the other human system variables as the bar (Fig. 2). The more resilient the human system is, the more pliant the bar and, therefore, the greater flexibility there is for maintaining balance. The ability to achieve sustainable development goals depends on two conditions: (1) the use of technologies that are compatible with the human systems variables and that make effective use of the environmental functions to provide human needs without violating the constraints; and (2) a social system that makes the choice of such technologies feasible. The fact that we have at present only an embryonic understanding of the environment and few effective levers over the human systems makes the pursuit of sustainable development a formidable task. This is not to justify inaction but to underline the need for reflective and collective action.

Those who are organizing for sustainable development need to address three questions: what are the goals of development (who is it for), sustainable over what area and for how long?

"Development" has to do with people. It involves people's goals, aspirations and ideas about how they think things should be. Development has been defined in a number of ways, including "economic growth", "industrialization", "increasing consumer choice", "increasing political participation", "empowerment of people", etc. Most definitions of development are perhaps understandably, oriented toward human systems. The debates in development have been about the nature of the process, the nature of the institutions, and the structures established to achieve the goals defined for development. In none of these definitions is the need for environmental balance directly implied. Development is viewed as a linear, unidimensional process. The environment, where considered at all, is seen as a passive factor of production, a passive dimension. Even grass-roots-oriented development models are only now starting to shed this linear perspective.

Environmentalists, on the other hand, uphold the cyclical character of the environment. There is, however, a tendency among environmentalists to view the environmental system as separate from, instead of encompassing, the human system. In this view, environmental conservation implied exclusion of humans. But, from the point of view of ecology, humans are similar to other predators. We are predators whose ability to affect the environment is much greater than most and it is precisely because of this greater power that humans have to be integrated into environmental management and planning.

Sustainable development requires the fusion of the linear people-oriented development model with the cyclical people-free environmental model. A successful synthesis should lead to a systems model of development, a dynamic interaction between the human and the environmental potential and constraints with the equilibrium point moving, depending on the balance between the two. It should also lead to a conservation model that recognizes and respects the fact that humans are part of .the environment. In the final analysis, sustainable development is a coming to terms with the fact that humans need nature and each other.

Figure 1

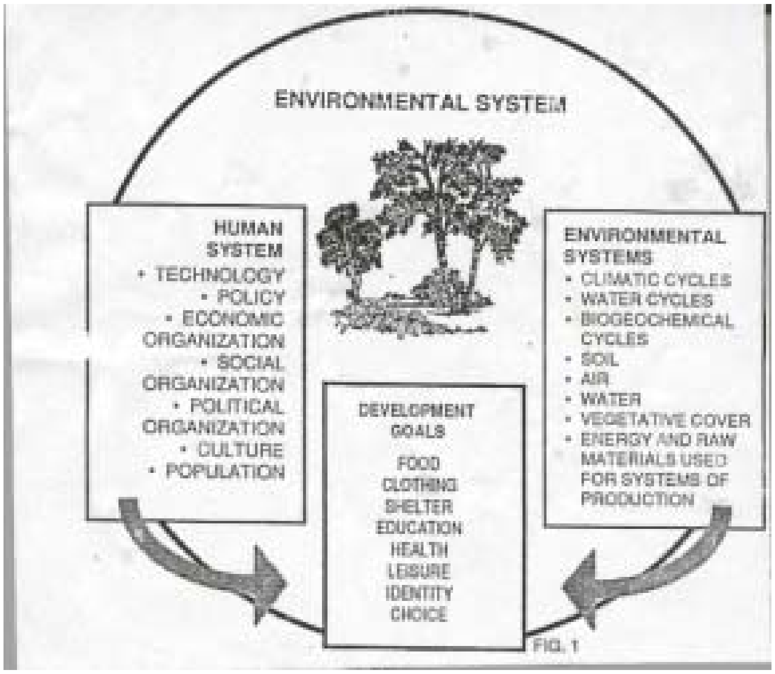

Figure 2